

Ryan Cho

Staff Full Stack Engineer

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PROFESSIONAL SUMMARY

Full stack engineer with 7+ years delivering production platforms across financial services and investment operations. Deep expertise in React, TypeScript, Python, and Node.js, architecting APIs, real-time messaging integrations, and event-driven microservices on AWS. Led front-end standards, web and chat channel integrations, and AI adoption in production workflows. Ships reliable demo-ready experiences, mentors engineers, and communicates complex architectural tradeoffs to cross-functional stakeholders.

WORK EXPERIENCE

Software Engineer

May/2022 - May/2026

Aspen Capital

Remote | 10th Floor, New York

- Defined **front-end architecture** and component standards for a **React and TypeScript** portfolio analytics platform serving **200+ internal users**, establishing shared design patterns that reduced duplicate UI work across three product squads.
- Led **end-to-end API design** in **Python and Node.js**, building RESTful and event-driven services that processed **15K+ daily transactions** with sub-second latency through caching and query optimization.
- Built **real-time messaging integrations** connecting web dashboards to **Slack** notification pipelines and webhook-driven alert workflows, improving operational response times for portfolio managers.
- Architected **microservices** with **message queues** for asynchronous trade reconciliation and reporting jobs, improving batch reliability and enabling independent service deployment cycles.
- Integrated **LLM-powered workflows** including streaming responses, tool calling, and retrieval-augmented generation into analyst tooling, accelerating research synthesis while maintaining audit-friendly output logging.
- Drove **AWS deployment patterns** using container orchestration, **CI/CD pipelines**, and infrastructure-as-code practices that shortened release cycles for critical production services.
- Implemented **authentication and authorization** controls across multi-tenant APIs, aligning with financial compliance requirements through role-based policy enforcement.
- Established **observability standards** with centralized logging, distributed tracing, and SLA dashboards that reduced mean time to detection for production incidents.
- Mentored **three junior and mid-level engineers** through code reviews, design discussions, and pairing sessions, raising delivery consistency and front-end testing coverage.

Technology Analyst, Incoming Technology Analyst, Technology Summer Analyst,

May/2019 - Apr/2022

BNP Paribas

Remote | Seventh Avenue, New York

- Developed **React and AngularJS** interfaces for capital markets analytics tools used by **global trading desks**, delivering accessible UI components aligned with enterprise design guidelines.
- Built **backend APIs** in **Python** supporting real-time pricing feeds and trade lifecycle workflows, integrating legacy systems through adapter services and message-driven pipelines.
- Optimized **relational and NoSQL database** schemas for regulatory reporting modules, applying indexing strategies and cache layers that improved query performance on large historical datasets.
- Designed **Twilio** and SMS notification hooks for operational alerts and client communication workflows within compliance constraints, enabling timely delivery across voice and text channels.
- Contributed to **microservices migration** initiatives decomposing monolithic batch processors into independently deployable services with **CI/CD** automation and containerized test environments.
- Partnered with business analysts and product owners to prototype **demo-ready UI flows** for stakeholder reviews, translating complex financial workflows into polished, navigable experiences.

- Enforced **testing practices** across unit, integration, and UI test suites, improving release confidence for monthly production deployments across multiple asset classes.
- Collaborated on **security hardening** including OAuth-based authentication flows and authorization policies for internal banking applications handling sensitive client data.

Software Engineering Intern

Jun/2018 - May/2019

Chexology

Remote | 17th Floor, New York

- Built **JavaScript and React** features for a restaurant check-tracking SaaS product, implementing responsive UI components consumed by **50+ venue partners** during pilot rollout.
- Developed **Node.js API endpoints** for order and billing workflows, integrating third-party payment callbacks with idempotent webhook handling.
- Wrote **unit and integration tests** for core service modules, contributing to improved test coverage before major feature releases.
- Assisted in **CSS and layout refinements** across customer-facing dashboards, improving mobile usability scores during user acceptance testing cycles.
- Participated in agile ceremonies and code reviews, gaining hands-on experience with **Git workflows** and continuous integration pipelines.

Web Developer

Sep/2017 - Nov/2017

Bayesquare Foundation

Remote | New York

- Developed **HTML, CSS, and JavaScript** pages for nonprofit fundraising campaigns, delivering mobile-responsive layouts that supported seasonal donation drives.
- Integrated **REST API** calls for donor management features, enabling real-time contribution tracking on public-facing web properties.
- Collaborated with stakeholders to implement accessibility-minded UI updates aligned with organizational branding guidelines.
- Maintained deployment scripts and performed content updates across staging and production environments during a **three-month** engagement.

EDUCATION

New York University

Bachelor of Arts (B.A.), Computer Science

2016-2020

SKILLS

Frontend: React, TypeScript, JavaScript, AngularJS, Next.js, HTML5, CSS3, Responsive UI, State Management, Figma Collaboration

Backend: Python, Node.js, REST APIs, Microservices, Event-Driven Architecture, Message Queues, Webhooks, LLM Integration, Streaming, Tool Calling, RAG

Databases: PostgreSQL, MongoDB, Redis, Query Optimization, Schema Design, Caching Strategies, NoSQL

Cloud & DevOps: AWS, Docker, Kubernetes, CI/CD, Terraform, Container Orchestration, Infrastructure as Code

Monitoring & Tools: Datadog, CloudWatch, Distributed Tracing, Centralized Logging, Slack, Twilio, Git, Jest, PyTest, Cypress

Workflow & PM: Agile, Scrum, Technical Leadership, Mentorship, Stakeholder Communication, Code Review, Prototyping, Cross-Functional Collaboration